PRITOM MAJUMDER

B.Sc. in Materials and Metallurgical Engineering
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LinkedIn | Research Portfolio | Undergraduate Thesis (PDF)

RESEARCH INTERESTS

- **Functional & Piezoelectric Ceramics**: Lead-free ferroelectrics (KNN-based systems), solid-state synthesis, dielectric and piezoelectric response, and low-temperature sintering routes.
- **Oxide Semiconductor Thin Films**: Doping-driven control of structural, optical, magnetic and electronic behaviour in transparent conducting oxides.
- **Photonic & Optoelectronic Materials**: Linking structure-property understanding to working devices: lasers, LEDs, photovoltaics and magnetic data storage.

EDUCATION

B.Sc. in Materials and Metallurgical Engineering **2021 – June 2026**
Bangladesh University of Engineering and Technology (BUET), Dhaka
CGPA: 3.64 / 4.00 **Merit position:** 15 of 59 in the graduating class

Relevant coursework: Advanced Ceramics; Photonics and Magnetic Materials; Electrical, Optical and Magnetic Properties of Materials; Nanostructured Materials & Thin Films; Characterisation of Materials; Crystallography and Structure of Materials; Phase Diagrams & Transformations; Mechanical Behaviour of Materials; Surface Engineering & Tribology; Corrosion & Degradation of Metals and Alloys.

PUBLICATIONS

Under peer review

1. **Pritom Majumder**, Md. Atik Faisal, Mehrab Rahman, Anabil Dey, and Md. Azizar Rahman. *"Structural, Optical, Magnetic, and Electronic Properties of Hydrothermally Grown Cu-Doped SnO_2 Thin Films: A Combined Experimental and First-Principles (DFT+U) Study."* Manuscript under review, 2026. **(First author)**

In preparation

2. **Pritom Majumder**, Anabil Dey, and Ahmed Sharif. *"Advancements in Cold Sintering of Piezoceramics: Flux Mediation, Microstructural Evolution, and Enhanced Dielectric Properties — A Review."* Manuscript in preparation. **(First author)**

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher · Department of MME, BUET **2025 – 2026**
"Investigation of Dielectric and Piezoelectric Properties of KNN with BCHfT Composite Ceramics" Supervisor: Prof. Dr. Ahmed Sharif

- Synthesized lead-free $K_{0.5}Na_{0.5}NbO_3$ (KNN) ceramics modified with $Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$ (BCHfT) by the conventional solid-state route, as an environmentally safer alternative to lead-based PZT.
- Characterized phase evolution and microstructure by XRD and FE-SEM; measured frequency- and temperature-dependent dielectric permittivity and loss, impedance spectroscopy, and the piezoelectric charge coefficient d_{33}.
- Identified KNN-1 wt% BCHfT as the optimum composition, giving the best balance of dielectric response and low loss together with the highest unpoled d_{33} measured in the series (15.5 pC/N).

SELECTED PROJECTS

Millsorp - Capstone Design Project · Team of 6, Department of MME, BUET **2025**

- Converted ferrous mill scale, a low-value steel-industry waste, into a chemically treated hematite-rich iron-oxide adsorbent for heavy-metal wastewater treatment.
- Designed a low-cost bench-scale filtration column achieving up to 95.8% removal of Cu, Cr and Ni from mixed-metal solutions; iron-oxide phases confirmed by XRD, metal removal quantified by AAS and UV–Vis.
- Presented the work to specialists at Abul Khair Steel, BCSIR and the Bangladesh Atomic Energy Commission, receiving positive technical feedback.

3D PotterBot - CAD Modelling & Finite Element Analysis · SolidWorks **2023**

- Modelled a paste-extrusion ceramic 3D printer as a fully-mated 118-part assembly; produced photorealistic renders, an exploded-view motion study, and a static FEA of the vertical support rod (≈15 mm peak deflection, strain and von Mises stress fields).

INDUSTRIAL TRAINING

Abul Khair Steel Melting Limited (AKSML) · Chittagong **March 2025**

- Studied electric arc furnace, ladle refining furnace and continuous casting operations, and rolling-mill practice using Quenching & Tempering Bar (QTB) technology for structural steel.
- Observed quality-control workflows spanning ICP-OES, XRF, handheld XRF and wet chemical analysis, together with process automation and real-time monitoring.

Bangladesh Industrial Technical Assistance Centre (BITAC) · Dhaka **March 2025**

- Hands-on training in precision machining (lathe, milling, CNC), TIG/arc/spot welding, heat treatment, foundry operations, CAD-CAM and industrial automation.

TECHNICAL SKILLS

- **Synthesis & Processing:** Solid-state ceramic synthesis; hydrothermal synthesis; sol-gel routes; sintering, uniaxial pressing and poling; solution casting.
- **Structural & Microstructural:** X-ray diffraction (XRD, X'Pert HighScore); FE-SEM and SEM; TEM including HRTEM, SAED and EDS; optical metallography; ImageJ grain analysis.
- **Electrical & Optical:** Impedance spectroscopy; frequency- and temperature-dependent dielectric measurement; d_{33} meter; cyclic voltammetry; I–V and Mott–Schottky analysis; UV–Vis spectroscopy; photoluminescence; FTIR.
- **Mechanical & Chemical Testing:** Tensile, Charpy impact and three-point flexural testing; Vickers, Rockwell and Brinell hardness; optical emission spectroscopy (OES); gravimetric and wet chemical analysis.
- **Software:** OriginPro; MATLAB; SolidWorks; ImageJ; X'Pert HighScore; C/C++; MS Office; Adobe Photoshop and Illustrator.

AWARDS & SCHOLARSHIPS

- **University Merit Scholarship**, Bangladesh University of Engineering and Technology (2023–24)
- **University Stipend Scholarship**, Bangladesh University of Engineering and Technology (2022-23 & 2025-26)
- **Letter of Appreciation**, Abul Khair Steel Melting Limited - for professionalism and work ethic during industrial training, 2025.

LANGUAGE PROFICIENCY

- **English:** IELTS Academic - Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.5), 2026.
- **Bangla:** Native.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Director of Administration**, Executive Committee, BUET Debating Club (BUETDC), 2025-26 :- previously Information Secretary. Organizing panel member for BUETDC Nationals 2024; finalist at the BUET Freshers' Debating Tournament; competed at the Inter-Faculty Tournament 2024 and the 6th GSCDC Nationals 2024.
- **Private Tutor**, intermediate-level students, 2022–2026 :- delivered regular one-to-one academic tuition alongside full-time undergraduate study.
- **Organizer**, Winter Fitness Competition 2025, BUET Gymnasium :- planned and ran the event in collaboration with the resident trainer.
- **Class Representative**, Department of MME, BUET, 2021–22 :- elected by classmates for both semesters of the first year.
- **Active Member & Blood Donor**, Badhon - a non-profit voluntary blood donation organization, 2022-present.

REFERENCES

Prof. Dr. Ahmed Sharif *Professor, PhD* Department of Materials and Metallurgical Engineering, BUET asharif@mme.buet.ac.bd +880 2 5516 7100, Ext. 7681 *Undergraduate Thesis Supervisor*	**Dr. Md. Moniruzzaman** *Professor* Department of Materials and Metallurgical Engineering, BUET mmoniruzzaman@mme.buet.ac.bd +880 2 5516 7100, Ext. 7605